Arithmetic and harmonic realizations of the Clebsch cubic

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Abstract

We determine the rational square class of Hitchin's degree-two incidence cover of harmonic cubics. Its function field is obtained by adjoining $\sqrt{5J_0}$, and on the Clebsch four-space $J_0 = 16\sigma_3^2$; the nonbranch restriction is therefore the constant golden torsor $\operatorname{Spec}\mathbf{Q}(\sqrt{5})$. The two configurations over the cubic xyz are conjugate over $\mathbf{Q}(\sqrt{5})$, and their exchanger has nonsquare spinor class after reduction modulo 11. Independently, the degree-six zonal harmonics on the ten icosahedral face axes contain the Clebsch four-space as their Petersen eigenspace, and their normalized spherical cubic restricts exactly as

$$\frac{1}{4\pi}\int_{S^2} F_y^3 = -\frac{784000}{1247103}\sigma_3(y).$$

Both results are expressed by the cubic σ_3 on the same four-space; no integral comparison between the two constructions is asserted.

Keywords. Clebsch configuration; Hitchin cover; icosahedral invariant; arithmetic double cover; spherical harmonics; Gaunt invariant.

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1 Introduction

Hitchin associated to a general harmonic cubic two icosahedral six-axis configurations [1, 2]. Retaining one of the two configurations gives a degree-two incidence cover of the projective seven-space of harmonic cubics. We determine its rational square class and then study a separate occurrence of the same Clebsch invariant in degree-six spherical harmonics.

Let H be the rational seven-space of harmonic cubics, let J_0 denote Hitchin's invariant sextic, and write

$$V = \{(y_1, \dots, y_5) : \textstyle\sum_i y_i = 0\}, \qquad \sigma_3(y) = \tfrac{1}{3}\textstyle\sum_i y_i^3$$

for the Clebsch four-space and its invariant cubic.

Theorem 1.1 (Arithmetic incidence cover)**.** *The function field of Hitchin's incidence cover is*

$$\mathbf{Q}(\mathbf{P}(H))(\sqrt{5J_0}), \qquad J_0|_V = 16\sigma_3^2.$$

Hence its restriction to $D(\sigma_3) \subset \mathbf{P}(V)$ is the constant golden torsor

$$D(\sigma_3) \times_{\operatorname{Spec}\mathbf{Q}} \operatorname{Spec}\mathbf{Q}(\sqrt{5}) \longrightarrow D(\sigma_3).$$

The fibre over xyz consists of two conjugate icosahedral configurations defined over $\mathbf{Q}(\sqrt{5})$. Their Galois exchanger has good reduction at 11, where its spinor class is the nontrivial element of

$$\mathrm{PGL}_2(11)/\mathrm{PSL}_2(11).$$

Theorem 1.2 (Degree-six harmonic restriction)**.** *Let u_{ij}, $1 \le i < j \le 5$, be the explicitly labelled unit axes through opposite icosahedral faces in* (1)*, and set*

$$F_y(\omega) = \sum_{i<j} (y_i + y_j) P_6(u_{ij} \cdot \omega), \qquad P_6(s) = \frac{231s^6 - 315s^4 + 105s^2 - 5}{16}.$$

The coefficient vectors $(y_i + y_j)_{i<j}$ form the Petersen (-2)-eigenspace, and their zonal harmonics give an injective copy of V in $\mathcal{H}_6$. With $\langle f, g\rangle = (4\pi)^{-1} \int_{S^2} fg$, the spherical cubic on this copy is

$$\frac{1}{4\pi} \int_{S^2} F_y(\omega)^3 \, d\omega = -\frac{784000}{1247103} \sigma_3(y).$$

The arithmetic theorem fixes the unknown constant in the square class of Hitchin's known cover by evaluating one golden fibre. Its statement at 11 concerns only that explicit fibre and exchanger. The abstract equation $z^2 = 5J_{\mathbf{Z}}$ is etale on $D(10J_{\mathbf{Z}})$, but its comparison with the geometric incidence variety is proved only after inverting an unspecified integer.

The harmonic theorem uses the same polynomial σ_3 on V, but it is logically independent of the incidence cover. The coefficient $-784000/1247103$ has denominator divisible by 11, and this paper constructs no common integral lattice from which the two realizations could be obtained by base change. The relation proved here is their common base four-space and invariant polynomial, not a specialization between them.

2 The rational incidence cover

The alternating group acts on the Clebsch four-space by permuting coordinates. If τ is odd, the rule $\kappa(y_i) = -y_{\tau(i)}$ defines a twisted involution. Put

$$A_R = R[y_1, \ldots, y_5]/(e_1), \qquad \Delta = \prod_{i<j} (y_i - y_j),$$

where $R = \mathbf{Z}[1/30]$. Nonmodular invariant theory gives

$$A_R^{A_5} = R[e_2, e_3, e_4, e_5, \Delta]/(\Delta^2 - \operatorname{disc}(e_2, e_3, e_4, e_5)).$$

$$\kappa(e_i) = (-1)^i e_i, \qquad \kappa(\Delta) = -\Delta.$$

Thus $e_3 = \sigma_3$ is the first nonconstant odd invariant. After inverting e_3^2, every odd element r is $e_3(r/e_3)$ with r/e_3 invariant, so the localized algebra is the direct sum of its even part and e_3 times that part. We use this observation only in characteristic zero. The classical binary-sextic presentation used by Hitchin is uniform in characteristic greater than 5 [2, 5]; modular invariant theory is not invoked here. Dye's finite-geometry theorem separately shows that Clebsch hexagons exist precisely when the characteristic is not 2 and 5 is a square [3, Theorem 1].

The threefold in the next statement originates with Mukai and Umemura; Hitchin's Grassmannian model supplies the incidence calculation used here [4, 2].

2.1 Proof of Theorem 1.1

Let $\mathcal{I}_{\mathbf{Q}}$ be the incidence variety of pairs $([f], U)$, where U is a Mukai–Umemura isotropic three-plane and $f \perp U$. Hitchin realizes the Mukai–Umemura threefold as the zero locus of the three skew forms on $\mathrm{Gr}(3, H)$. The top Chern number of the dual universal bundle is 2, so a general f is orthogonal to two isotropic three-planes [2, §§4–5]. The incidence variety is a projective bundle over the integral Mukai–Umemura threefold, hence is integral; its generic function field is therefore a quadratic extension.

Square class from the branch divisor. Write $K = \mathbf{Q}(\mathbf{P}(H))$. In characteristic zero every quadratic extension of K has the form $K(\sqrt{d})$. The prime divisors at which d has odd valuation are exactly the branch divisors. Hitchin identifies the branch hypersurface with the irreducible invariant sextic $J_0 = 0$ [1, §§9–10]. Consequently

$$\operatorname{div}(d/J_0) = 2D$$

for a divisor D on $\mathbf{P}(H)$. The class of D is two-torsion. Since $\operatorname{Pic}(\mathbf{P}(H)) = \mathbf{Z}$, D is principal. Thus

$$d = cJ_0 g^2$$

for some $g \in K^\times$ and a unique $c \in \mathbf{Q}^\times/\mathbf{Q}^{\times 2}$. The incidence extension is therefore $K(\sqrt{cJ_0})$.

Determining the constant from one fibre. Hitchin computes

$$J_0|_{\mathbf{P}(V)} = 16\sigma_3^2$$

and exhibits over $f = xyz$ the two ordered icosahedra with coordinates in the roots of $t^2 - t - 1$ [1, §§3 and 9–10]. Their residue algebra is $\mathbf{Q}(\sqrt{5})$. At this point $J_0(f) = (4\sigma_3(f))^2$ is a nonzero rational square, so the fibre of $K(\sqrt{cJ_0})$ has residue algebra $\mathbf{Q}(\sqrt{c})$. Hence $c = 5$ in $\mathbf{Q}^\times/\mathbf{Q}^{\times 2}$.

Restriction to the Clebsch chart. On $D(\sigma_3) \subset \mathbf{P}(V)$, the odd generator divided by $4\sigma_3$ has square 5. This gives the asserted product with $\operatorname{Spec}\mathbf{Q}(\sqrt{5})$.

Integral boundary. Choose a full lattice $\Lambda \subset H$ and an integer $a \neq 0$ such that $J_{\mathbf{Z}} = a^2 J_0$ has integral coefficients. The algebra

$$\mathcal{O}_{\mathbf{P}(\Lambda)} \oplus \mathcal{O}_{\mathbf{P}(\Lambda)}(-3), \qquad z^2 = 5J_{\mathbf{Z}},$$

is finite locally free of degree two over $\mathbf{Z}$ and has the function field of Theorem 1.1. On $D(10J_{\mathbf{Z}})$ it is finite etale. The primes 2 and 5 are forced bad primes for an ordinary two-sheet interpretation: in characteristic 2 the equation is inseparable, while in characteristic 5 it is generically $z^2 = 0$.

This abstract integral algebra does not by itself give an integral incidence theorem. The constructions in the cited Mukai–Umemura and Hitchin sources are over $\mathbf{C}$, and the available binary-sextic formulae justify their classical invariant presentation only after inverting 30. Spreading out the rational incidence isomorphism gives some $\mathbf{Z}[1/N]$, with $2, 3, 5 \mid N$ after enlarging N, but neither the sources nor the present equations determine the other prime divisors or prove that there are none. Consequently every finite-field icosahedral-incidence interpretation in this paper remains restricted to an unspecified cofinite set of primes. The formula for the abstract quadratic algebra itself has no such extra exception.

3 Arithmetic specialization

The arithmetic exchange is already visible on the cubic xyz. Its two icosahedral six-axis configurations are Galois conjugate over $\mathbf{Q}(\sqrt{5})$; modulo 11 they are rational, and the Galois exchange has nontrivial spinor class.

3.1 The golden fibre

Let $t^2 - t - 1 = 0$, and put

$$I_t = \{[0 : t : 1], [0 : t : -1], [1 : 0 : t], [-1 : 0 : t], [t : -1 : 0], [-t : -1 : 0]\} \subset \mathbf{P}^2.$$

These six projective points are its axes through opposite vertices. Thus an icosahedral configuration here means the unordered six-set I_t; the incidence cover remembers which of the two conjugate six-sets lies over the cubic.

Proposition 3.1 (Golden fibre)**.** *Over a field of characteristic different from* 2 *and* 5*, the points of* I_t *lie on* $xyz = 0$*, have a common nonzero norm, have normalized squared inner product* $1/5$*, and form a six-arc. The two configurations are conjugate geometric configurations defined over*

$$\mathbf{Q}[t]/(t^2 - t - 1) = \mathbf{Q}(\sqrt{5})$$

and are exchanged by its nontrivial automorphism. In Hitchin's normalized cover of ordered icosahedra they are the complete fibre over $[xyz]$*.*

Proof. Every displayed vector has one zero coordinate. Since $t^2 = t + 1$, its squared norm is $t^2 + 1 = t + 2$. The inner product of two distinct representatives is $\pm t$, while

$$(t + 2)^2 = 5(t + 1) = 5t^2.$$

This gives the normalized squared inner product $1/5$. The twenty three-point determinants are $\pm 2t$ or $\pm 2t^2$, so none vanishes under the stated hypotheses. The final assertion uses Hitchin's classification of the two icosahedral sets on the orthogonal-plane cubic. □

3.2 The exchanger

Set

$$R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

The relation $1 - t = -1/t$ gives, by substitution,

$$R(I_t) = I_{1-t}, \qquad R^2 = \mathrm{diag}(1, -1, -1), \qquad R^4 = 1.$$

Moreover $R(xyz) = -xyz$: it fixes the projective cubic and exchanges its two ordered icosahedra.

3.3 Reduction modulo eleven

The roots of $t^2 - t - 1$ in $\mathbf{F}_{11}$ are 4 and 8. For the displayed equations for I_t and R therefore reduce directly to two rational configurations I_4, I_8 exchanged by R.

Proposition 3.2 (Spinor specialization)**.** *The reduction of* R *represents the nontrivial element of*

$$T_{11} = \mathrm{PGL}_2(11)/\,\mathrm{PSL}_2(11).$$

Proof. For $Q(x,y,z)=x^2+y^2+z^2$, reflection in a vector v has spinor class $Q(v)$. On the yz-plane,

$$R = s_{e_2}s_{e_2-e_3},$$

so

$$\theta(R)=Q(e_2)Q(e_2-e_3)=2 \quad \text{in } \mathbf{F}_{11}^{\times}/\mathbf{F}_{11}^{\times 2}.$$

The class of 2 is nonsquare modulo 11. The standard identifications

$$\mathrm{SO}_3(11)\cong \mathrm{PGL}_2(11), \qquad \Omega_3(11)\cong \mathrm{PSL}_2(11)$$

identify the spinor quotient with T_{11}, proving the claim. $\square$

Integral boundary. Over a finite field of characteristic different from 2 and 5, the abstract quadratic cover gives

$$\#X_f(\mathbf{F}_q)=1+\chi_q(5J(f)).$$

Its geometric interpretation over finite fields is proved only away from an unspecified finite set of primes. This global comparison is separate from the good reduction at 11 of the explicit golden fibre and matrix R; we do not replace the finite exceptional set by $p\neq 2,5$.

4 The degree-six harmonic realization

Put $\phi=(1+\sqrt{5})/2$. The following vectors have squared norm 3; their projective classes are the ten axes through opposite faces of an icosahedron:

$$\begin{array}{llll} v_{12}=(-1,-1,-1), & v_{13}=(-1,-1,1), & v_{14}=(-1,1,-1), & v_{15}=(-1,1,1),\\ v_{34}=(0,-\phi^{-1},-\phi), & v_{25}=(0,-\phi^{-1},\phi), & v_{45}=(-\phi^{-1},-\phi,0), & \\ v_{23}=(-\phi^{-1},\phi,0), & v_{35}=(-\phi,0,-\phi^{-1}), & v_{24}=(\phi,0,-\phi^{-1}). & \end{array} \tag{1}$$

We use the unit representatives $u_{ij}=v_{ij}/\sqrt{3}$. Two labels are adjacent when the pairs are disjoint. This is the Petersen graph $KG(5,2)$, and the labeling is geometric because, for distinct axes,

$$(u_{ij}\cdot u_{kl})^2=\begin{cases}5/9, & \{i,j\}\cap\{k,l\}=\varnothing,\\ 1/9, & \text{otherwise.}\end{cases}$$

The zonal degree-six harmonics define

$$L:\mathbb{R}^{10}\longrightarrow \mathcal{H}_6, \qquad L(a)(\omega)=\sum_{i<j} a_{ij}P_6(u_{ij}\cdot\omega),$$

where

$$P_6(s)=\frac{1}{16}\left(231s^6-315s^4+105s^2-5\right).$$

We equip $\mathcal{H}_6$ with the normalized spherical inner product

$$\langle f,g\rangle=\frac{1}{4\pi}\int_{S^2} f(\omega)g(\omega)\,d\omega.$$

4.1 Proof of Theorem 1.2

Let A be the Petersen adjacency matrix in the labeling (1). Direct evaluation gives the reproducing-kernel matrix

$$K_{ef} = P_6(u_e \cdot u_f) = \frac{1}{243}(196I + 47J - 112A).$$

The addition theorem for spherical harmonics gives

$$G_{ef} = \langle P_6(u_e \cdot -), P_6(u_f \cdot -)\rangle = \frac{1}{13}K_{ef}.$$

Thus G, not K, is the Gram matrix for the stated inner product. Its eigenvalues on $\mathbb{R}^{10} \simeq \mathbf{1} \oplus V_4 \oplus V_5$ are respectively

$$\frac{110}{1053}, \qquad \frac{140}{1053}, \qquad \frac{28}{1053}.$$

They are positive, so L is injective.

The Petersen eigenvalues are $3, -2, 1$ on $\mathbf{1}, V_4, V_5$. For $a_{ij} = y_i + y_j$ with $\sum_i y_i = 0$,

$$(Aa)_{ij} = \sum_{\{k,l\}\cap\{i,j\}=\varnothing} (y_k + y_l) = -2(y_i + y_j).$$

These vectors form the four-dimensional (-2)-eigenspace.

The cubic integral is A_5-invariant on V_4. The invariant cubic line is generated by σ_3, so one nonzero value determines the scalar. At $y = (4, -1, -1, -1, -1)$, exact spherical moments give

$$\frac{1}{4\pi}\int F_y^2 = \frac{2800}{351}, \qquad \frac{1}{4\pi}\int F_y^3 = -\frac{15680000}{1247103},$$

while $\sigma_3(y) = 20$. This proves the coefficient in Theorem 1.2. The moments follow by expanding P_6 and applying

$$\frac{1}{4\pi}\int_{S^2} \omega_1^{2a}\omega_2^{2b}\omega_3^{2c}\, d\omega = \frac{(2a-1)!!(2b-1)!!(2c-1)!!}{(2a+2b+2c+1)!!};$$

monomials with an odd exponent integrate to zero.

Remark 4.1. In the orthonormal Condon–Shortley convention, write $F = \sum_{m=-6}^{6} q_{6m}Y_{6m}$ and

$$W_6(F) = \sum_{m_1+m_2+m_3=0} \begin{pmatrix} 6 & 6 & 6 \\ m_1 & m_2 & m_3 \end{pmatrix} q_{6m_1}q_{6m_2}q_{6m_3}.$$

The Gaunt formula and $\left(\begin{smallmatrix} 6 & 6 & 6 \\ 0 & 0 & 0 \end{smallmatrix}\right) = -20/\sqrt{46189}$ give

$$\int_{S^2} F^3\, d\omega = -\frac{130}{\sqrt{3553\pi}}\, W_6(F).$$

This is the standard unnormalized degree-six bond-order invariant [6, 7, 8].

5 Reproducibility and data availability

The square-class argument and both specialization propositions are proved in the text from Hitchin's incidence calculation and the displayed equations. A paper-local exact-arithmetic bundle independently checks the golden configurations, the exchanger, and its spinor decomposition.

A second paper-local bundle reconstructs the ten axes over $\mathbf{Q}(\sqrt{5})$, distinguishes the reproducing kernel from the normalized spherical Gram matrix, verifies the Petersen labeling, and evaluates the quadratic and cubic moments. Its independent replay uses a separate dense polynomial representation and checks the Wigner $3j$ conversion. Neither calculation uses floating-point arithmetic.

The claim/evidence map, exact programs, canonical certificates, checksums, toolchain requirements, and aggregate replay command are included in the `verification` directory distributed with the paper. The aggregate runner also checks the four theorem-like statements against their frozen identity and requires a warning-free manuscript build.

6 Conclusion

On the Clebsch chart, Hitchin's branch sextic becomes $16\sigma_3^2$. Evaluating one nonbranch fibre determines the remaining rational square class and turns the restricted incidence cover into the constant golden torsor. The explicit conjugate configurations also show that this arithmetic statement has a well-defined specialization at 11 without requiring a global integral incidence model there.

The degree-six face-axis construction gives a second exact role for σ_3: it is the restriction of the normalized spherical cubic to the Petersen four-space. This shares a four-space and invariant polynomial with the arithmetic restriction, but no canonical integral comparison between the two constructions is presently known.

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